Convolutional Layer

A Convolutional Layer is made up of a large number of convolution filters, each of which is a template that determines whether a particular local feature is present in an image. A convolutional filter relies on a very simple operation, called a convolution, which basically amounts to repeatedly multiplying matrix elements and then adding the result.

$$\text{Original Image} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{bmatrix}.$$

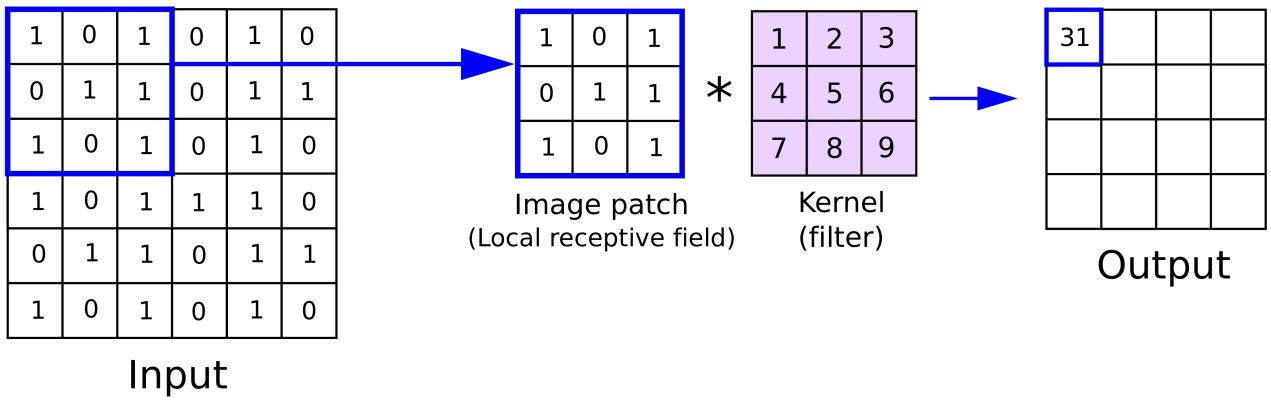
Now consider a 2×2 filter of the form

$$\text{Convolution Filter} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}.$$

When we *convolve* the image with the filter, we get the result[7]

$$\text{Convolved Image} = \begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}.$$

How Convolutional Layers work?



- **A small matrix** called a **filter** (**kernel**) slides over the input image to extract important features.
- At each position, the filter performs element-wise multiplication with the image patch.
- The multiplied values are summed together to produce a single output value.
- This operation is repeated across the entire image using a defined stride.
- The result is a new matrix called a feature map, which highlights detected patterns.
- Multiple filters are applied to capture different features such as edges, textures and shapes.

- The process preserves spatial relationships while reducing the number of learnable parameters compared to fully connected layers.
- Padding can be used to control output size and prevent loss of border information.